免示教拍照点标定工具使用说明 | Instruction Manual for Calibration Tool of Teach-Free Photographing Points

一、标定工具确认 | 1. Calibration Tool Confirmation

确认计算机D盘下是否有免示教标定工具：CalibrationTool

Check if the calibration tool without teaching is available under the D drive of the computer:CalibrationTool

二、旧数据清理 | 2. Cleaning of Old Data

重新标定时，确保D:\Bioyond\VisionServer\Locator\Config\Test\Long or Wide下的标定数据已经清空

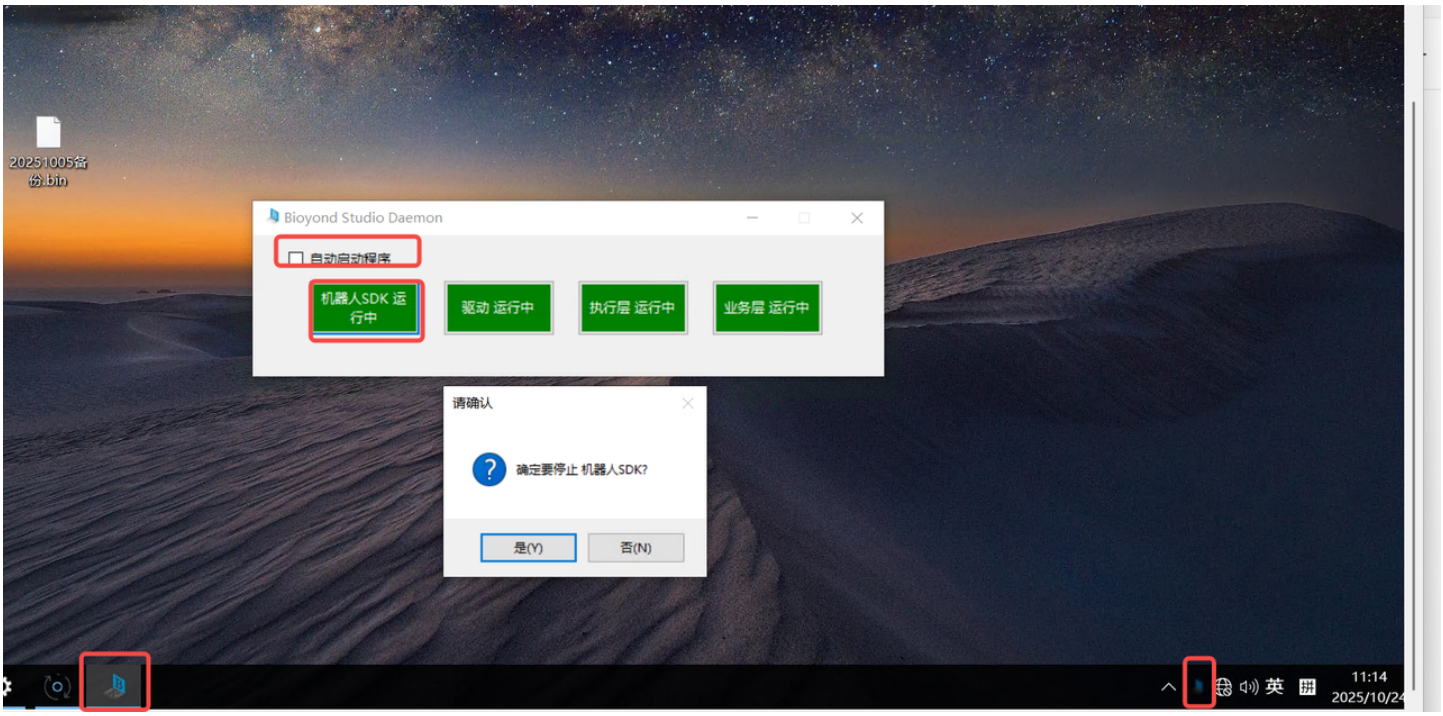
When recalibrating, ensure that the calibration data under D:\Bioyond\VisionServer \ Locator \ Config \ Test \ Long or Wide has been cleared

三、重新标定数据 | 3. Recalibrate Data

1、关闭设备的机器人SDK | 1. Turn off the device's robot SDK

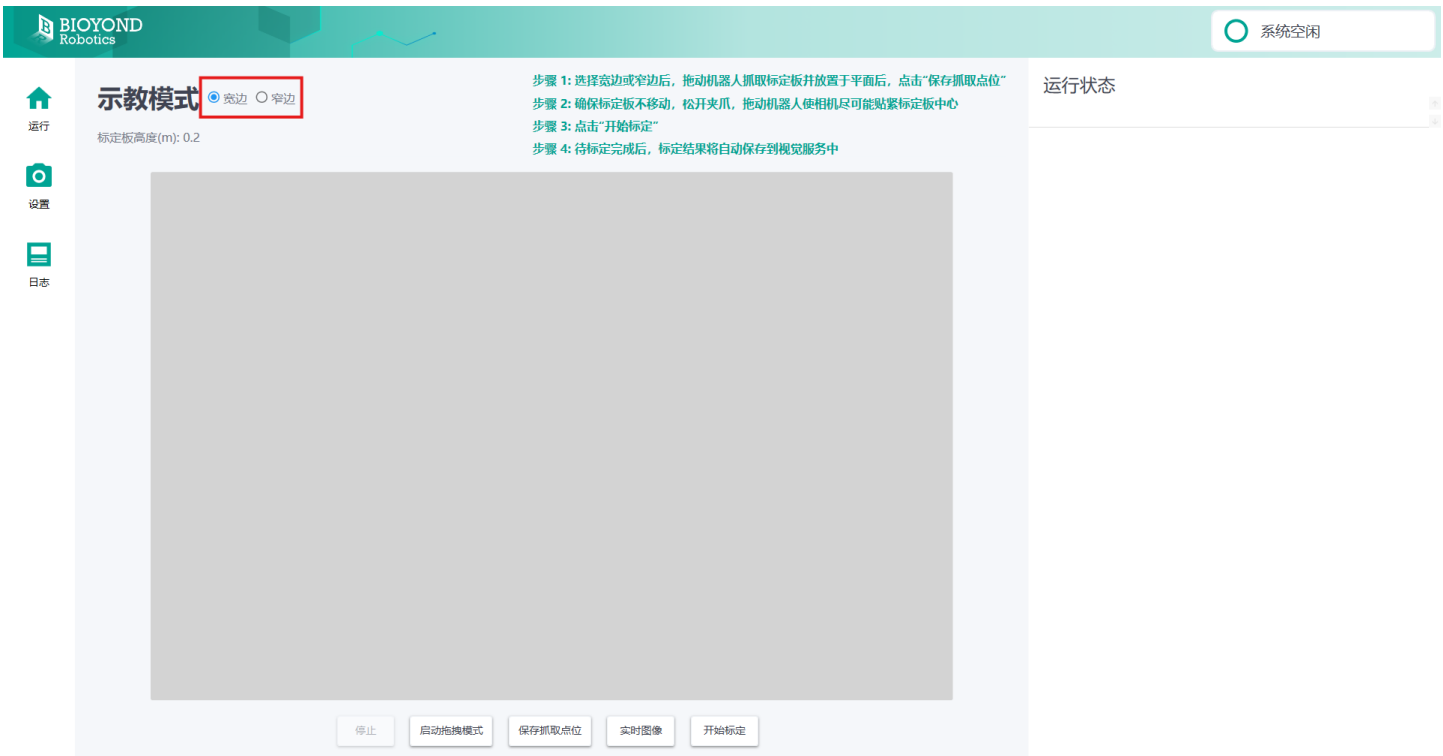
关闭机器人SDK（免示教标定工具会调用机器人，需要关闭Diana系统的占用）

Turn off the robot SDK (the teaching-free calibration tool will call the robot, so the occupation of the Diana system needs to be released)



2、打开免示教标定工具，按照以下步骤进行标定 | 2. Open the teach-free calibration tool and perform calibration according to the following steps

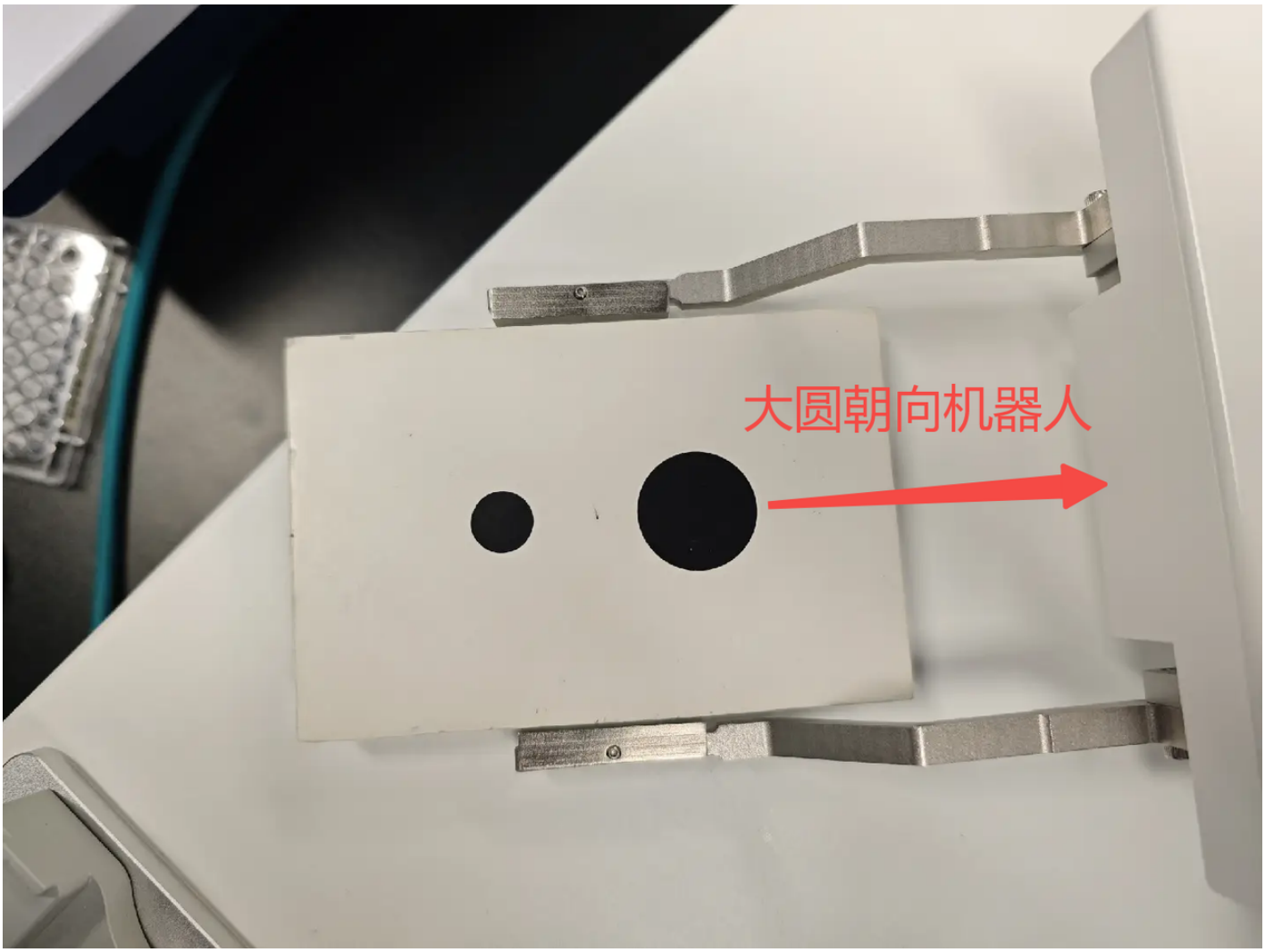
步骤一：选择宽边或窄边的抓取方式 | Step 1: Select the gripping method for wide or narrow edges



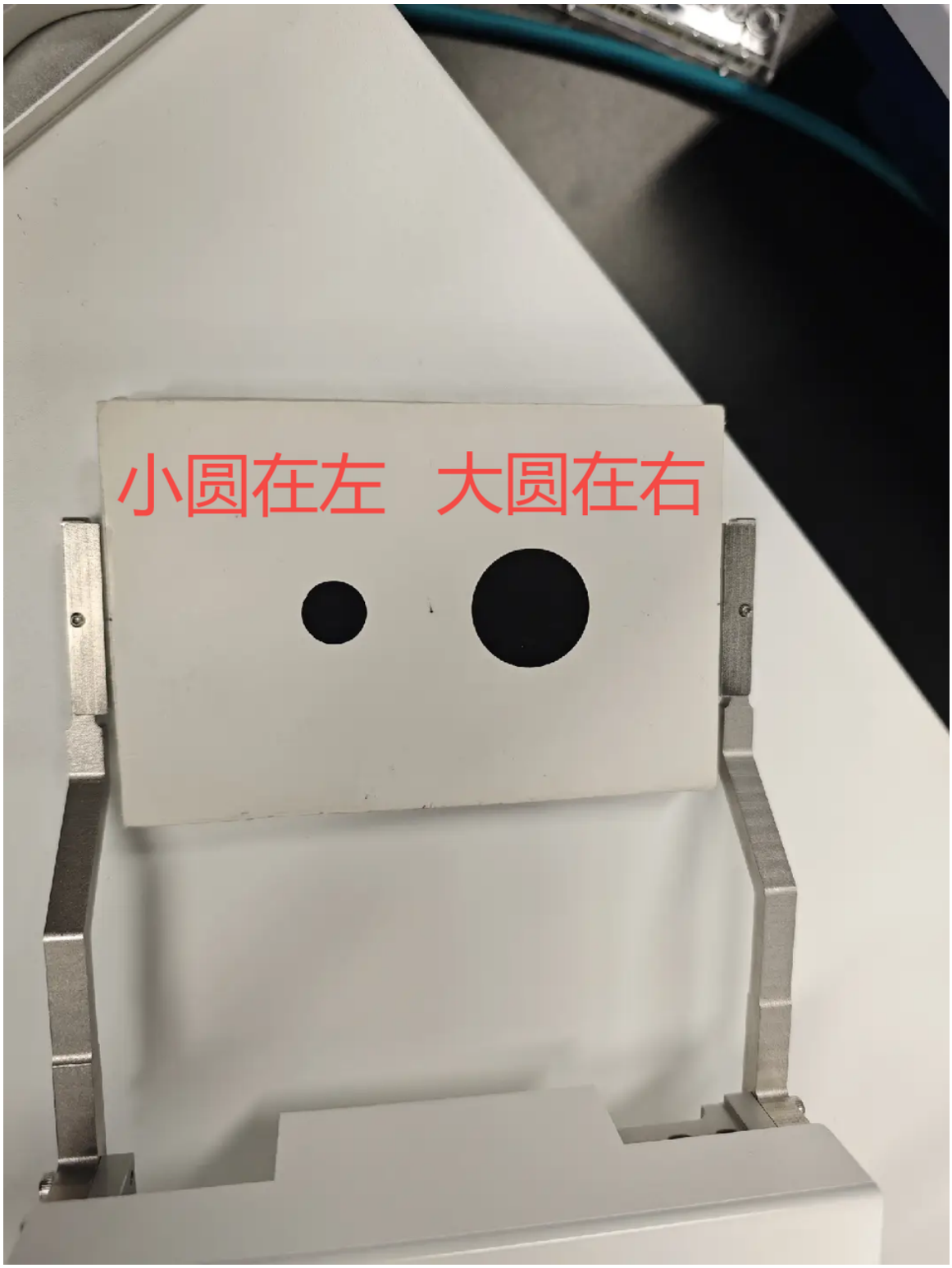
点击“启动拖拽模式”，拖动机器人抓取标定板并放置于平面上（放置详情参考下图），点击“**保存抓取点位**”。

Click "Start Drag Mode", drag the robot to grab the calibration board and place it on the plane (see the placement details in the figure below), click " **Save Grab Points** ".

宽边抓取方式： | Wide-edge grasping method:



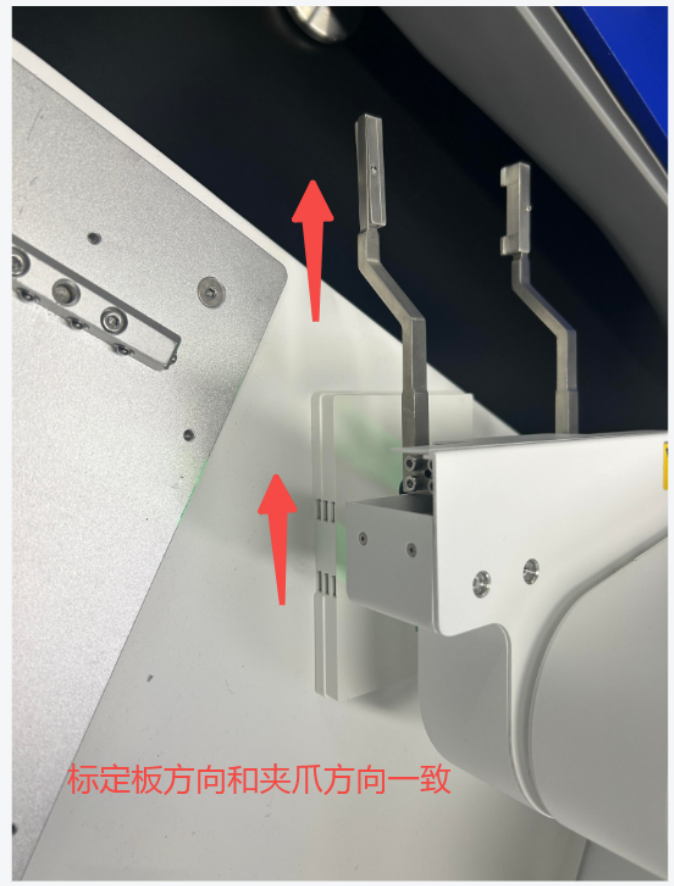
窄边抓取方式： | Narrow Edge Grasping Method:



步骤二：第一次标定 | Step 2: First Calibration

确保标定板不移动，松开夹爪，拖动机器人使相机尽可能贴紧标定板中心。此时保证夹爪方向和标定板夹持方向一致如图所示

Ensure that the calibration board does not move, release the gripper, and drag the robot so that the camera is as close as possible to the center of the calibration board. At this time, ensure that the direction of the gripper is consistent with the clamping direction of the calibration board as shown in the figure.



宽边抓取方式标定时，拖动机器人的相机在大圆的上方

When calibrating the wide-edge grasping method, drag the robot's camera above the large circle

窄边抓取方式标定时，拖动机器人的相机在大圆和小圆的中心，并稍稍偏向于机器人方向

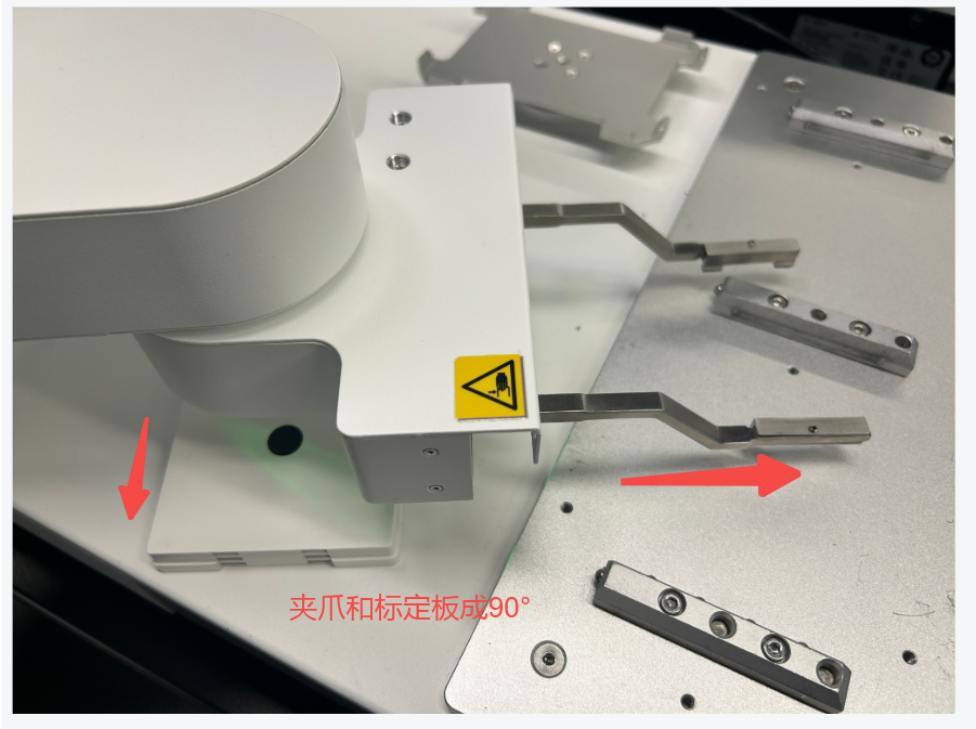
When calibrating the narrow-edge grasping method, drag the robot's camera to the centers of the large and small circles, and slightly bias it towards the robot direction

步骤三：点击“开始标定”，待标定完成后，标定结果将自动保存到视觉服务中。 | Step 3: Click "Start Calibration". After calibration is completed, the calibration results will be automatically saved to the vision service.

步骤四：第二次标定 | Step 4:Second Calibration

在确保标定板不动的情况下，改变J4轴的角度，使其尽量贴近标定板的中心，使标定板和夹爪成90°

While ensuring that the calibration plate remains stationary, change the angle of the J4 axis to make it as close as possible to the center of the calibration plate, so that the calibration plate and the gripper form a 90° angle



步骤五：更换宽边/窄边标定 | Step 5: Replace wide/narrow edge calibration

如第一次是宽边抓取方式，此时更换为窄边抓取方式，重复步骤一、二、三、四，可以进行多个拍照点位的标定（通常标定二至四个点位，每个点位相差约90度）。

If the first time is the wide-edge grasping method, then change to the narrow-edge grasping method at this time, repeat steps one, two, three, and four, and calibration of multiple photographing points can be performed (usually calibrating two to four points, with each point differing by approximately 90 degrees).